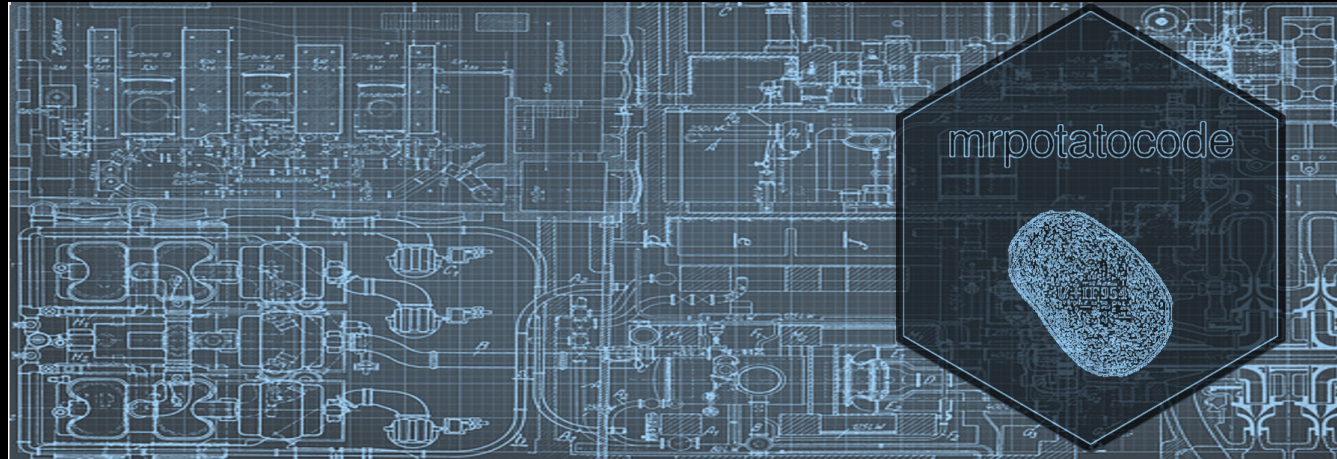




Introduction to Data Access and Storage

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Session 03

Essential Techniques:

Aggregation Functions

Subqueries

Temporary Tables & CTEs

Datetime Functions

Essential Techniques:

Aggregation Functions

Subqueries

Temporary Tables & CTEs

Datetime Functions

Aggregation Functions

GROUP BY

COUNT

SUM & AVG

MIN & MAX

Arithmetic

HAVING

Aggregation Functions

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HAVING

GROUP BY

- Aggregations generally require a *group*
- `GROUP BY` is mandatory any time a column aside from the one being aggregated is present

For example, a query wanting to know the number of days in each month:

```
SELECT  
COUNT(days)  
  ,months  
  
FROM calendar  
GROUP BY months
```

- `GROUP BY` comes after a `WHERE` clause
- Remember: with all aggregation functions, values will be calculated *per group*

Aggregation Functions

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HAVING

COUNT

- `COUNT` performs counts of any given column or set of columns
- `COUNT(*)` provides a count of rows in a given table
 - no `GROUP BY` is required here
- Multiple counts are treated as separate columns
 - e.g. counting the number of days and months in a year

```
SELECT  
COUNT(days)  
,COUNT(DISTINCT months)  
,years  
  
FROM calendar  
GROUP BY years
```


COUNT

- When `COUNT` is combined with `DISTINCT`, only unique values are counted
 - `SELECT COUNT(DISTINCT product_id) ...` might produce a different value than `SELECT COUNT(product_id) ...` depending on the context of the table

vendor_id	product_id	market_date
1	1	2026-01-01
1	1	2026-01-02
1	2	2026-01-01
1	2	2026-01-02
1	2	2026-01-03
1	3	2026-01-02
1	4	2026-01-02

vendor_id	count_of_distinct_products
1	4

vendor_id	count_of_products
1	7

COUNT

(COUNT live coding)

Aggregation Functions

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HAVING

SUM & AVG

SUM

- `SUM` performs the sum total of any numeric column
 - Be wary, SQLite may be more permissive for columns with numbers; it's best practice to coerce (`CAST`) these values into numbers before summing to be certain of their validity
 - e.g. `SUM(CAST(column1 AS INTEGER)) AS column1`
- `SUM` can accommodate multiple columns using the plus `+` operator
 - e.g. `SUM(column1 + column2)`
- Thinking about `SUM` and `COUNT` combined (i.e. a rolling total)? We'll get to that in the next session!

SUM & AVG

AVG

- **AVG** performs the average of any numeric column
- Like **SUM**, it can accommodate multiple columns
 - we can also use other mathematical operations for **SUM** and **AVG**, like $-$, $*$, $/$, $\%$ (i.e. modulo, not percent), etc

SUM & AVG

AVG

Watch out! Don't average an average column when using `GROUP BY`

Imagine, `market_avg_temp` stored in the `market_date_info` table:

market_day	market_avg_temp
Saturday	36
Sunday	33
Wednesday	25
Saturday	28
Sunday	31
Wednesday	31
Saturday	29

If we `GROUP BY market_day`, we can produce an average for each day of the week:

market_day	dow_market_avg_temp
Saturday	31
Sunday	32
Wednesday	28

Avoid averaging `dow_market_avg_temp` to get an `overall_market_avg_temp`:

actual_avg	avg_of_avgs
30.42857	30.33333

SUM & AVG

(SUM & AVG live coding)

Aggregation Functions

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HAVING

MIN & MAX

- `MIN` takes the single minimum value of a given column; `MAX` takes the maximum
- Be wary of combining `MIN` & `MAX` with other aggregating functions like `SUM` or `AVG`
- What do we think happens when `MIN` is performed on a string? Error? Something else? What about `MAX`?

MIN & MAX

(MIN & MAX live coding)

Aggregation Functions

GROUP BY

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MIN & MAX

Arithmetic

HAVING

Arithmetic

- SQL can perform many basic (and some complex) calculations
 - addition, subtraction, multiplication, division, power, etc
 - geometric/trigonometric functions sin, cos, tan, degrees, radians, etc
- These calculations can also be combined inside aggregation functions
 - e.g. multiplication inside a SUM ...`SUM(quantity * cost)` would create a column like total_spent per group
- SQL is similar to other programming languages in its ability to handle floating point values
- **Because columns are type specific, how would we perform integer division on two numbers?**

Arithmetic

(Arithmetic live coding)

Aggregation Functions

GROUP BY

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MIN & MAX

Arithmetic

HAVING

HAVING

- **WHERE** clauses filter rows *before* an aggregation occurs
 - ...so **HAVING** clauses allow us to filter rows *after* an aggregation is calculated
- **HAVING** clauses come after **GROUP BY**, but before **ORDER BY**
- **HAVING** clauses only filter aggregated calculations
 - you can have both **WHERE** and **HAVING**
 - they are not interchangeable

HAVING

(HAVING live coding)

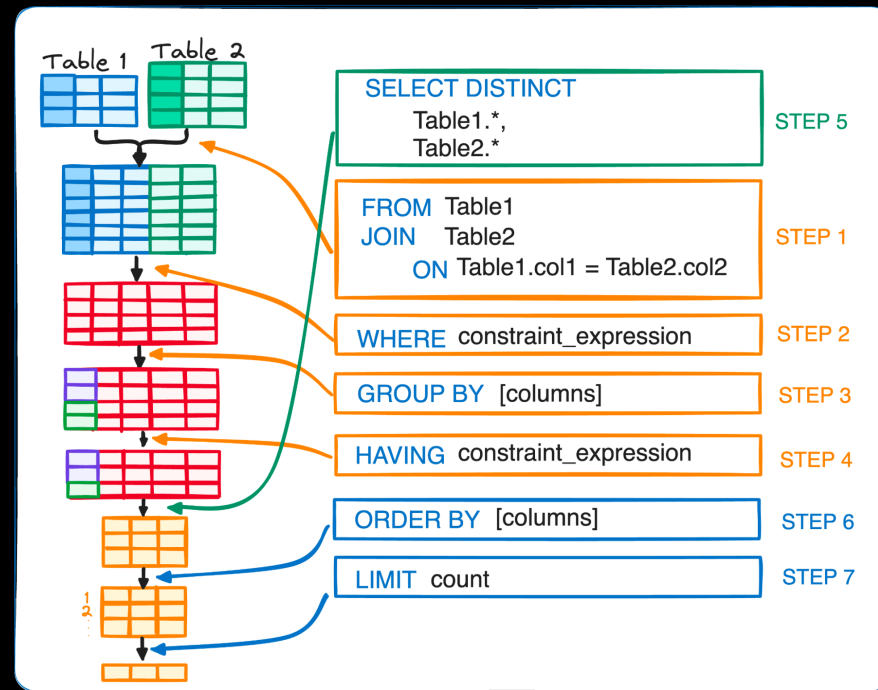


Image: Joseph Ferrer, KDnuggets

Essential Techniques:

Aggregation Functions

Subqueries

Temporary Tables & CTEs

Datetime Functions

Subqueries

- SQL allows us to query the results of another query
 - We call this a subquery
- Subqueries can be used in both **JOIN** and **WHERE** clauses
 - In the case of **JOIN** (or **FROM**):
 - you want the subquery to add columns to your output
 - you are using a subquery because you are joining/selecting complex criteria that require manipulation
 - it is often the case that you are joining two or more tables within a subquery to another table

SELECT *	<i>outer query:</i>
FROM	<i>everything from the subquery</i>
(	
SELECT	<i>subquery:</i>
vendor_id	<i>ordering each vendor's</i>
,market_date	<i>items by price</i>
,product_id	
,original_price	
, ROW_NUMBER() OVER (PARTITION BY vendor_id	
ORDER BY original_price DESC) AS price_rank	
FROM vendor_inventory	
) x	
WHERE x.price_rank = 1	<i>outer query:</i>
	<i>filtered by price_rank</i>

Subqueries

- Subqueries can be used in both **JOIN** and **WHERE** clauses
 - In the case of **WHERE**:
 - you want to filter results
 - you are using a subquery because it is simpler than joining and filtering the columns otherwise
 - it's important to note: *you can only return a single column* in your subquery
 - **Why do we think this is?**

```
SELECT
market_date
,customer_id
,vendor_id
,SUM(quantity*cost_per_quantity) as total_spent
FROM customer_purchases

WHERE market_date IN
(
    SELECT market_date
    FROM market_date_info
    WHERE market_rain_flag = 1
)
```

outer query:
each sale that's occurred

subquery:
all days it has rained

- In a subquery, all columns need to be uniquely named
- Subqueries can usually be run for testing purposes by highlighting them, IDE dependent

Subqueries

(Subqueries live coding)

Essential Techniques:

Aggregation Functions

Subqueries

Temporary Tables & CTEs

Datetime Functions

Temporary Tables & CTEs

Temporary Tables

CTEs

Temporary Tables & CTEs

Temporary Tables

CTEs

Temporary ("temp") Tables

- Table objects created on the fly
- Automatically saved to a reserved `temp` schema
- Accessible across SQL queries in the same session
- Cleared from memory when SQL is closed (or the server connection is terminated)
- Temporary tables can be chained in the same query
 - You can place one temporary table into another
- Must be dropped (deleted from memory) to recreate them with the same name
- Some older versions of SQL don't allow temporary tables
- They are *fantastic placeholders*
 - **What scenarios can we think of where a temporary table would be particularly useful?** 🗨️💬 **Think, Pair, Share**

Temporary Tables

(Temporary Table live coding)

Temporary Tables & CTEs

Temporary Tables

CTEs

Common Table Expressions (CTEs)

- Similar to temporary tables
 - CTEs were developed *before* temp tables
 - Some SQL versions/flavours (especially much older ones) might not support temp tables, so CTEs are an important skill
- Instantiated query results created on the fly
 - Utilize the `WITH` command
 - Many RDBMs require a semicolon terminating the `WITH` clause
 - Multiple CTEs don't use more than one `WITH` clause, but rather follow one another with a comma
 - Need to be written *before* the final `SELECT` statement
- Sometimes easier than a subquery
 - If subqueries become overly complex, they can be harder to read
- Stored in memory
- Limited to your current query window only

Common Table Expressions (CTEs)

(CTEs live coding)

Essential Techniques:

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Subqueries

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Datetime Functions

Datetime Functions

Formats

'NOW'

STRFTIME

Adding Dates

Difference between Dates

Datetime Functions

Formats

'NOW'

STRFTIME

Adding Dates

Difference between Dates

Formats

- Date formats vary widely in SQL databases
 - A general rule of thumb when working with multiple date fields is to force them all into a similar format
 - This may seem obvious, but different source systems may write dates different in SQL DBs
- It is not uncommon to store date values as integers YYYYMMDD to increase optimization and decrease storage size
- Manipulating dates varies by flavour
- SQLite is *less* flexible with dates, requiring all dates to either be:
 - "YYYY-MM-DD" *strings*
 - Julian Day *fractions*
 - Seconds from Unix Time *integers*

Datetime Functions

Formats

'NOW'

STRFTIME

Adding Dates

Difference between Dates

'NOW' (or GETDATE()) or DATE, flavour dependent)

- These functions (there are actually more of them) get the current date and time
 - Some will return UTC time if requested (this can be useful) e.g. `GETUTCDATE()`
- When combined with other Datetime functions, this can serve as a dynamic value
 - e.g. "yesterday", "last year", and so on
- SQLite uses `DATE()`, `DATETIME()`, `TIME()` (without any arguments) *or* `DATE('now')`

'NOW'

('NOW' live coding)

Datetime Functions

Formats

'NOW'

STRFTIME

Adding Dates

Difference between Dates

STRFTIME

- `STRFTIME` converts DATE and DATETIME values into different formats
- `STRFTIME` also allows you to extract specific "dateparts"
 - e.g. `SELECT STRFTIME( '%Y', 'NOW' )`
- The first argument of `STRFTIME` is flexible — you can specify more than one datepart at a time *and* any formatting
 - e.g. `SELECT STRFTIME( '%Y-%m', 'NOW' )` would return 2026-07

STRFTIME

- `STRFTIME` also allows modification to date dynamically
 - e.g. `SELECT STRFTIME('%Y-%m-%d', '2026-07-30', 'start of month')`
 - **How do we go about subtracting dates rather than adding them?**
- Modifiers include:
 - +/- N years/months/days/hours/minutes/seconds
 - start of year/month/day
 - weekday
- Be mindful: because outcome is a *string*, modification should be done within the `STRFTIME` argument to ensure it is correct
- Some flavours have built in convenience dateparts, like `YEAR`, `MONTH`, etc that make extracting values a bit easier

STRFTIME

(STRFTIME live coding)

Datetime Functions

Formats

'NOW'

STRFTIME

Adding Dates

Difference between Dates

Adding Dates (sometimes DATEADD or DATE_ADD, flavour dependent)

- SQLite supports two means of adding increments of time to a date:
 - `STRFTIME` as mentioned previously
 - Using `DATE`
 - e.g. `SELECT DATE('2026-07-30', 'start of month')`
- Both of these methods allow you to chain modifiers
 - e.g. `SELECT DATE('2026-07-30', 'start of month', '-1 day')`

What do we see as the difference between these?

- This syntax is fairly unique to SQLite, but is conceptually the same, so briefly I will touch on `DATEADD`
 - Generally, we specify a datepart, add/subtract a value, and the date

Datetime Functions

Formats

'NOW'

STRFTIME

Adding Dates

Difference between Dates

Difference between Dates (an extension of STRFTIME or DATEDIFF, flavour dependent)

- The difference between dates can vary in complexity
- We can use `STRFTIME`, subtracting the two dates from one another, using '%s' as our unit
 - e.g. `SELECT (STRFTIME("%s", Date1) - Date2) / {increment, e.g. 3600.0 for hours, 60.0 for minutes, etc}`
 - Be sure to include .0 for float precision: `ROUND` or `CAST` to integer if desired
 - `STRFTIME` works well for calculating months and years
 - e.g., months until next total solar eclipse 🌑🌞 `SELECT STRFTIME('%m', '2026-08-12') - STRFTIME('%m', 'NOW')`
- We can also use `JULIANDAY`:
 - Julian Days are *fractional* by nature and result in a difference of days
 - e.g., difference in hours `SELECT CAST((JULIANDAY(Date1) - JULIANDAY(Date2) * 24) AS INT)`
- This syntax is also fairly unique to SQLite, but is conceptually the same, so briefly I will touch on `DATEDIFF`
 - Generally, we specify a datepart, startdate, and enddate

Datetime Functions

(Adding Dates and Difference between Dates live coding)

